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LECTURE 2:

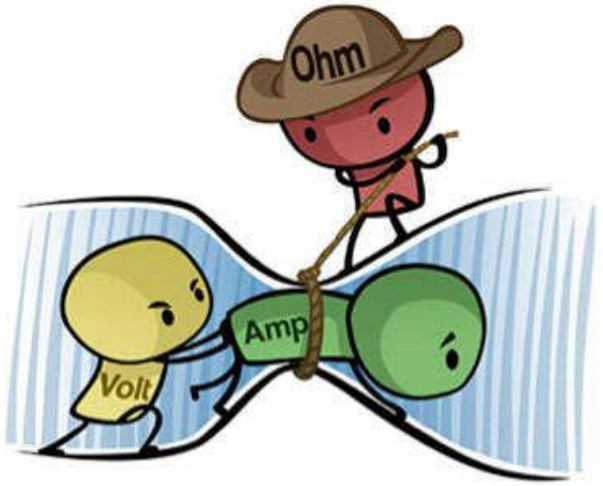
SOLDERING

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Recap Of Lecture 1

Circuit Rules

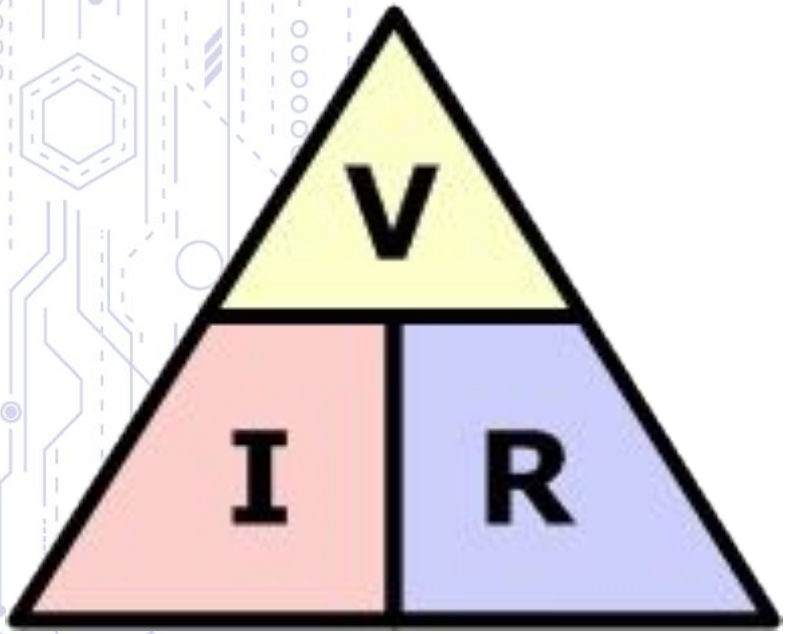


- Electrons flow from HIGH potential to LOW potential
- Voltage (V): Electric potential difference between two points; it drives the current through the circuit: volts (V)
- Current (I): The flow of electric charge through a conductor: amps (A)
- Resistance (R): The opposition to the flow of current in a material: ohms (Ω)

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Recap Of Lecture 1

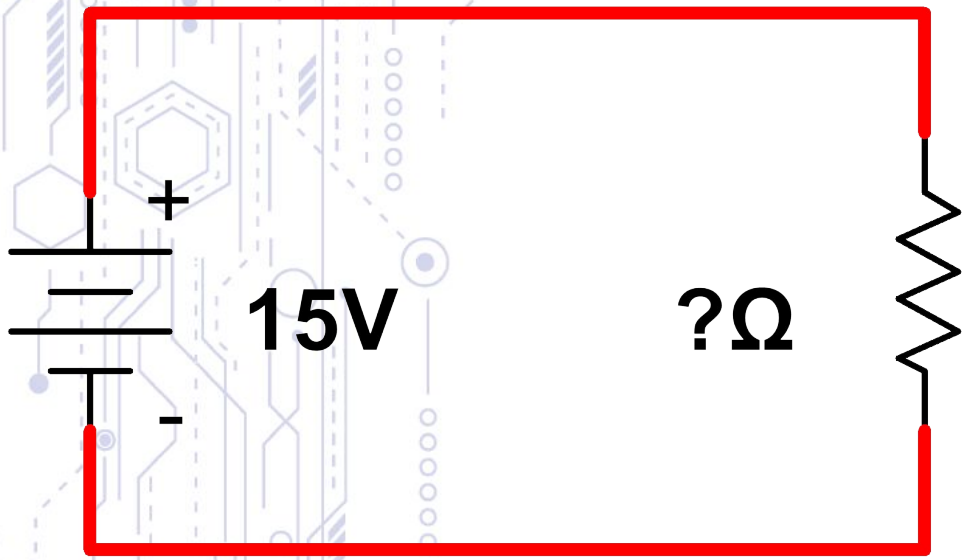
Ohm’s law



- **Ohm’s law: V = IR**
- **From knowing any two variables, you can solve for the third**

Recap Of Lecture 1

Ohm's law



Comprehension Check:

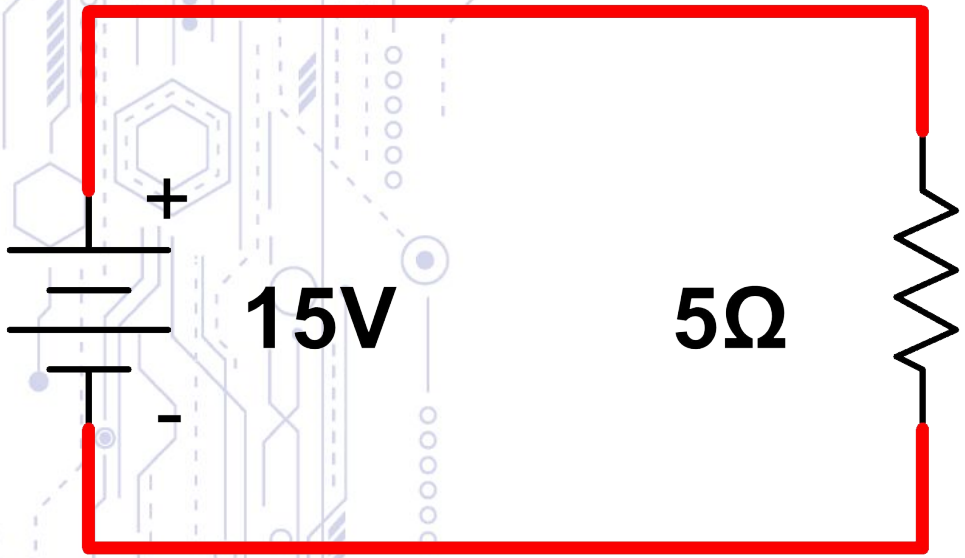
Across two nodes, there is a 15V difference.

There are 3A of current flowing. How much resistance is between the two nodes?

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Recap Of Lecture 1

Ohm's law



Comprehension Check:

Across two nodes, there is a 15V difference.

There are 3A of current flowing. How much resistance is between the two nodes?

15V = (3A) * (**5Ω**)

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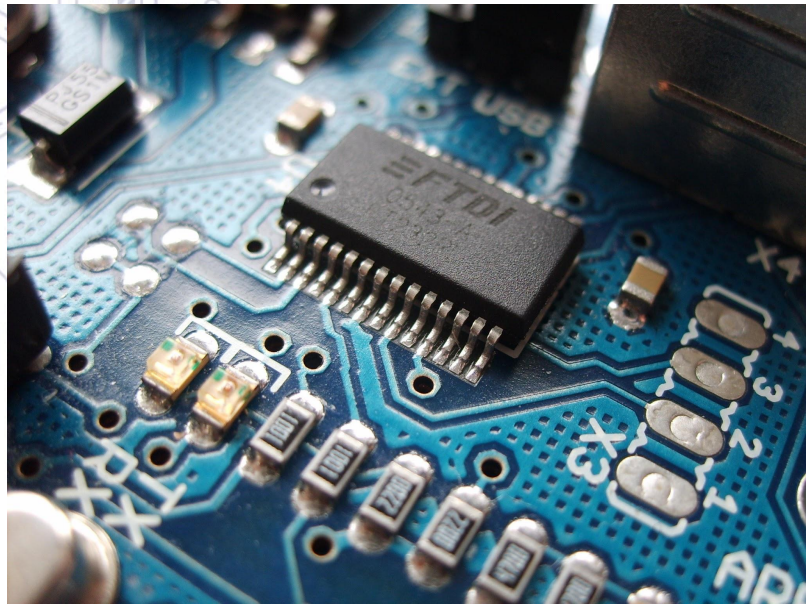
Solder Safety

Lab Tips

- The tip is EXTREMELY HOT!! Return it to the stand!
- Wear eye protection and keep the workspace uncluttered
- Use fume extraction; keep your face out of the plume
- No food or drinks; wash your hands after handling solder

Breadboard or Solder?

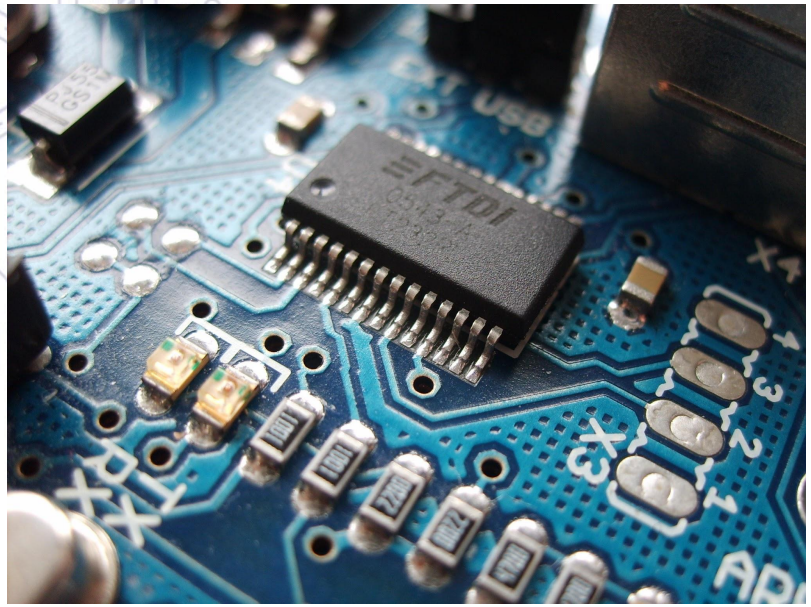
How to Choose



- **Breadboard: fast, reversible, and ideal for testing**
- **Soldered PCB: compact, durable, and electrically reliable**
- **Debug the circuit first, then commit it to hardware**
- **Rework is possible, but every heat cycle adds risk**

Why Solder?

Durability



- A solder joint creates both an electrical path and a mechanical bond
- Good joints survive movement, vibration, and repeated use
- Poor joints cause intermittent faults... the hardest kind to debug

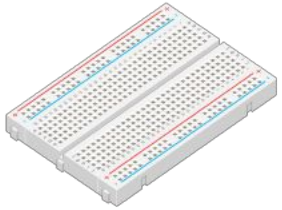
Types of Connections

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Simple to System Level

Level 1:

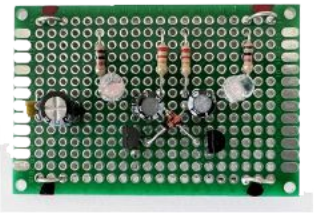
Breadboard



Used for early testing, experimentation, and debugging circuits

Level 2:

Perfboard/Stripboard



Used for simple prototypes, one-off projects, and hobby builds.

Level 3:

PCB Assembly (Board Level)



Components mounted and soldered to copper pads on PCB. This is the backbone of almost all modern electronics

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Types of Connections

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Simple to System Level

Level 4:

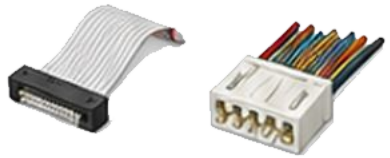
Package Internal Connections



Internal electrical connections inside every IC. Silicon connected to package using wire bonding or flip-chip bonding

Level 5:

Connectors/Cables



Board to board connections used to connect PCB's. Mechanical + Electrical connections

Level 6:

System Level Connections



Internal modules and components connect via cables, connectors, and mechanical interfaces rather than solder.

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How to Solder

Technique



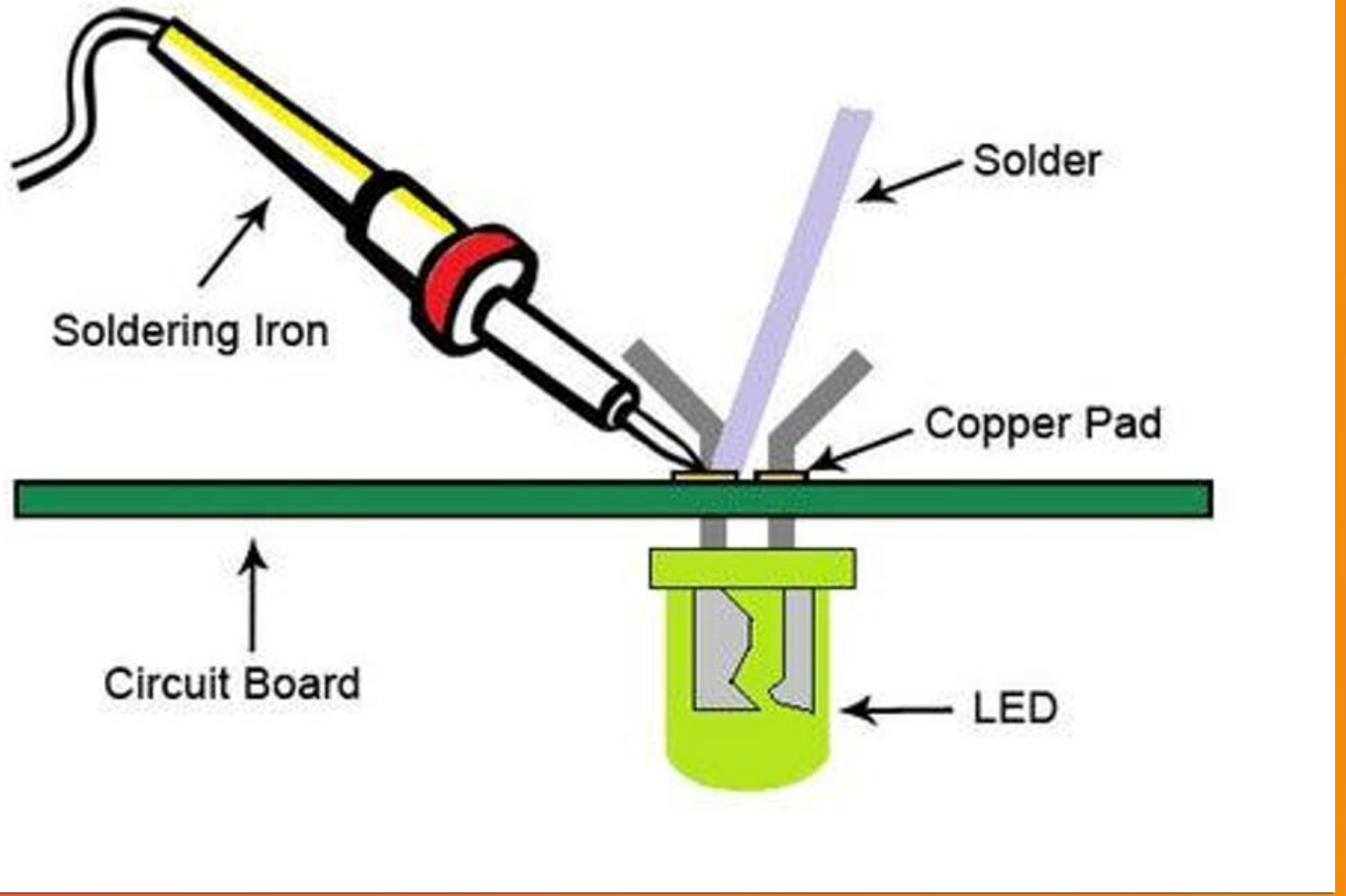
- Heat the pad and component lead together
- Feed solder into the heated joint.. NOT onto the iron
- Flux removes oxides so molten solder can wet the metal
- The cooled solder provides electrical and mechanical continuity

How to Solder

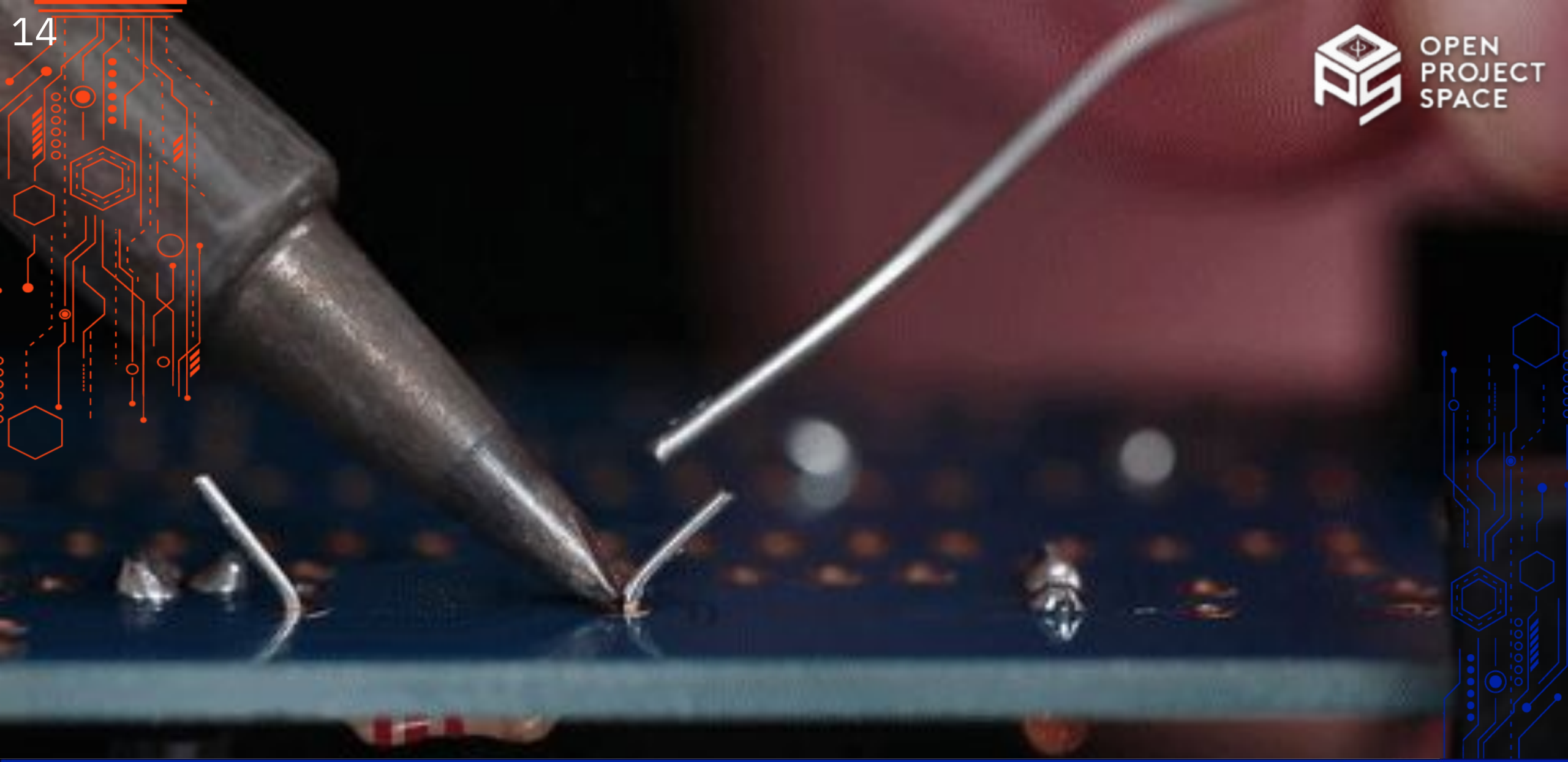
Tips

- 1) Clean and tin the tip
- 2) Touch the tip to both pad and lead
- 3) Feed a small amount of solder into the joint
- 4) Remove the solder, then remove the iron
- 5) Hold still while the joint cools

How to Solder



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Soldering Demonstration:

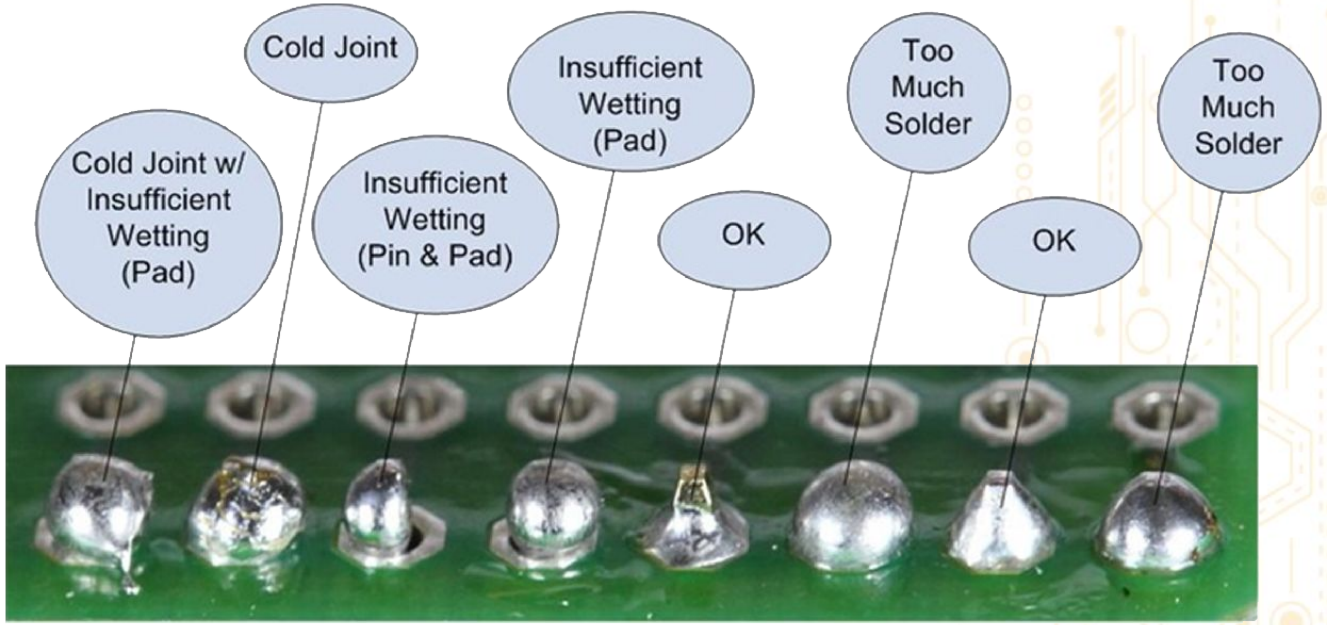
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Common Mistakes

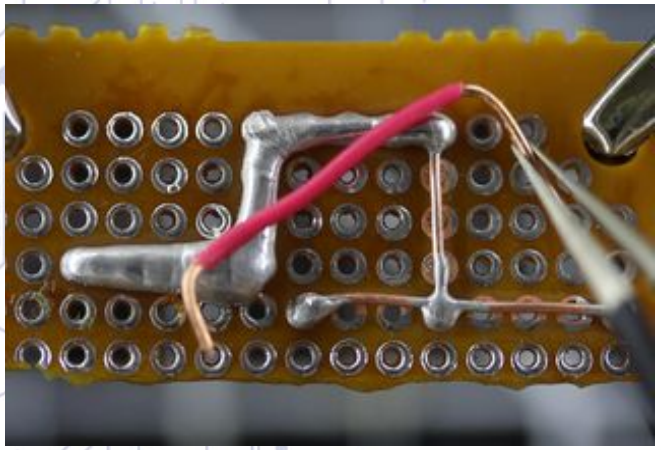
Hershey's Kiss

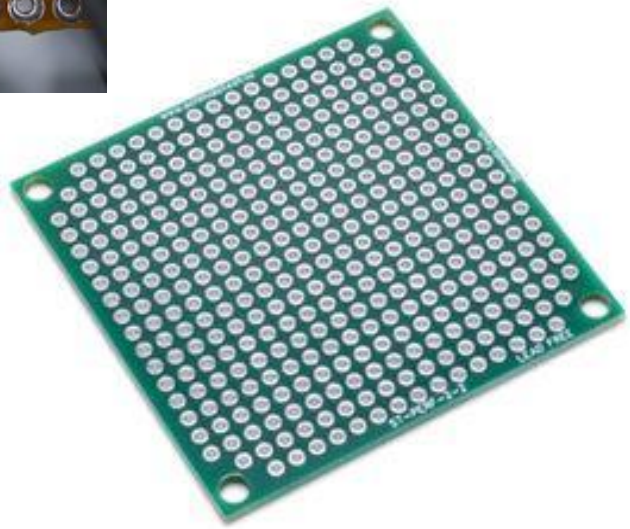




A good joint is smooth, wetted to pad and lead, and uses only enough solder

Perfboarding

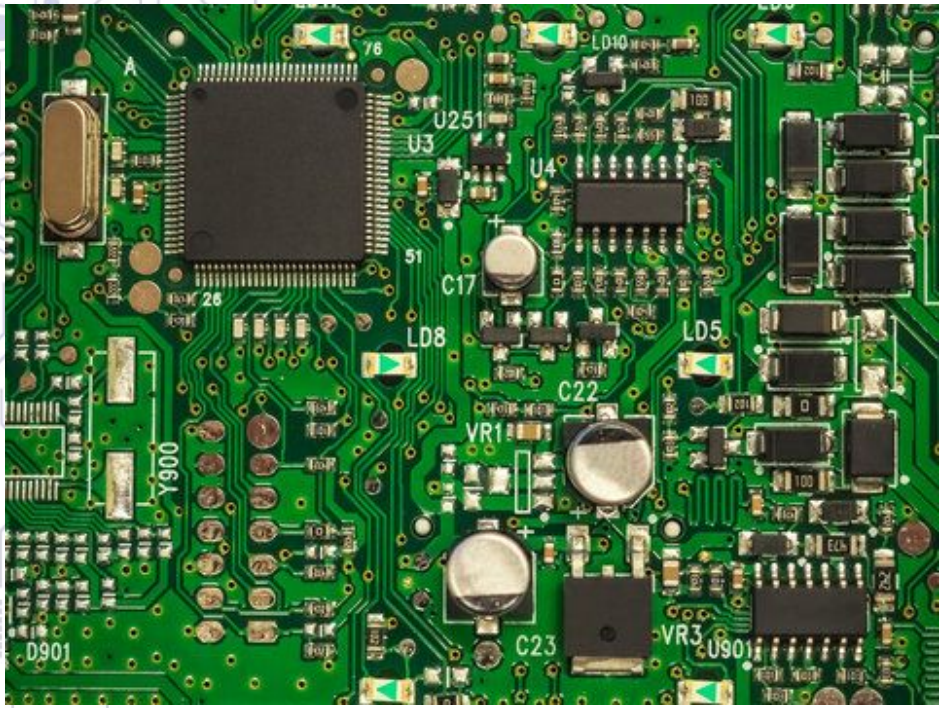




- Place components through a regular grid of holes
- Plan the component layout before soldering
- Create connections with leads, wire, or solder bridges
- Check continuity and shorts as the circuit grows

Printed Circuit Boards

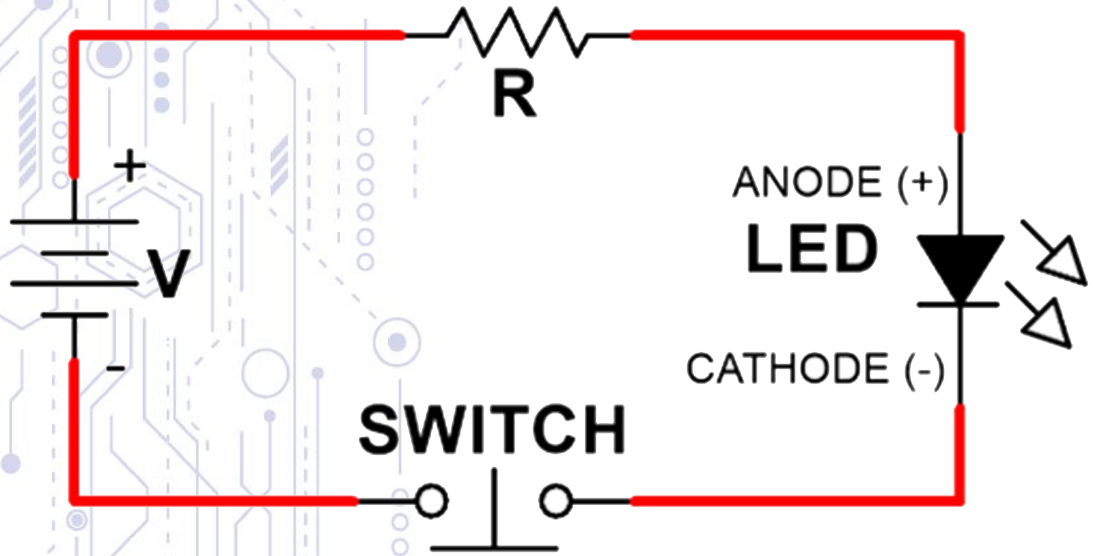
PCBs



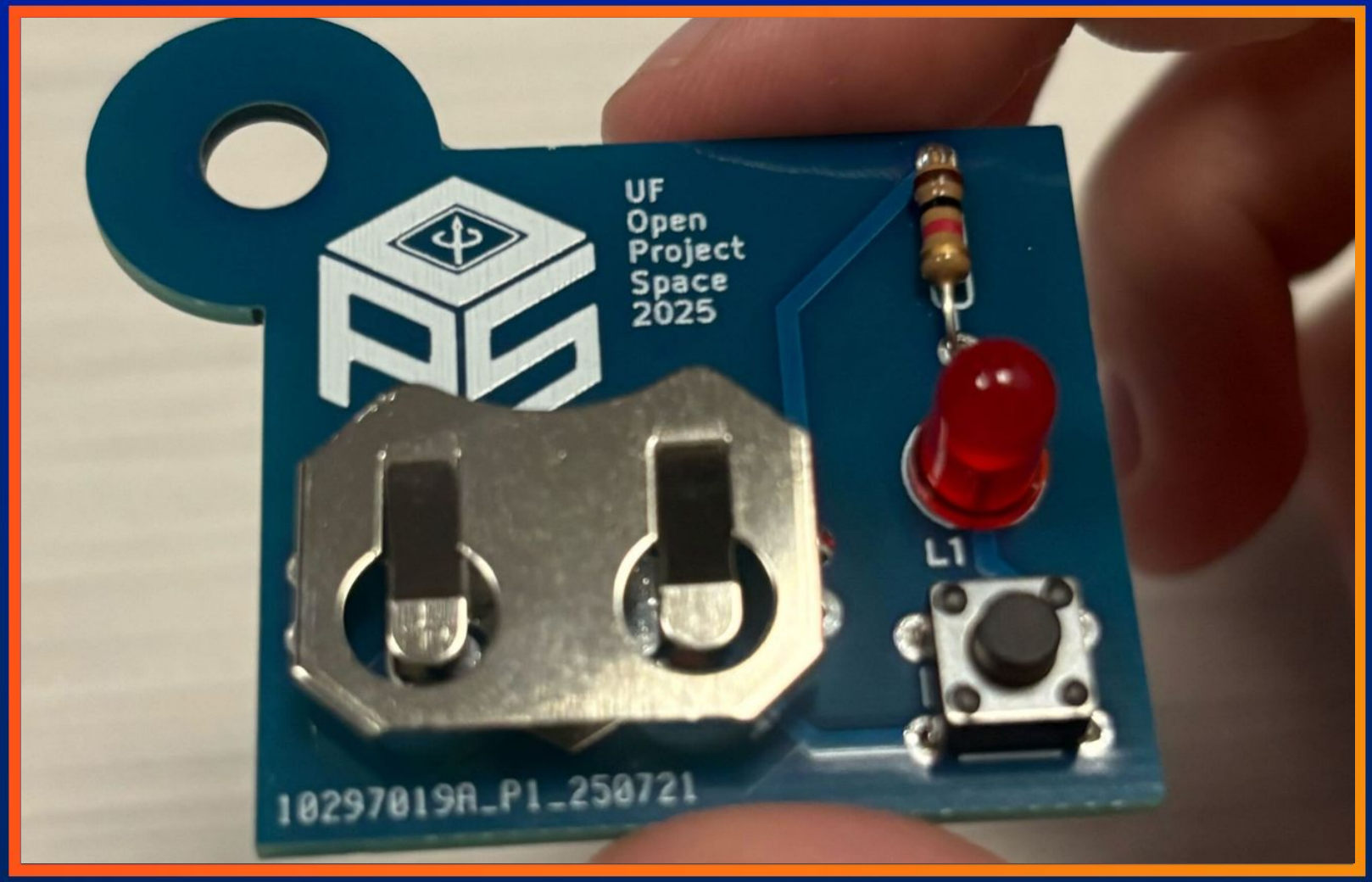
- Copper traces replace most point-to-point wiring
- Pads locate and connect component leads
- Silkscreen communicates placement and orientation
- Solder mask helps prevent unintended solder bridges

Circuit Diagram

LED Circuit

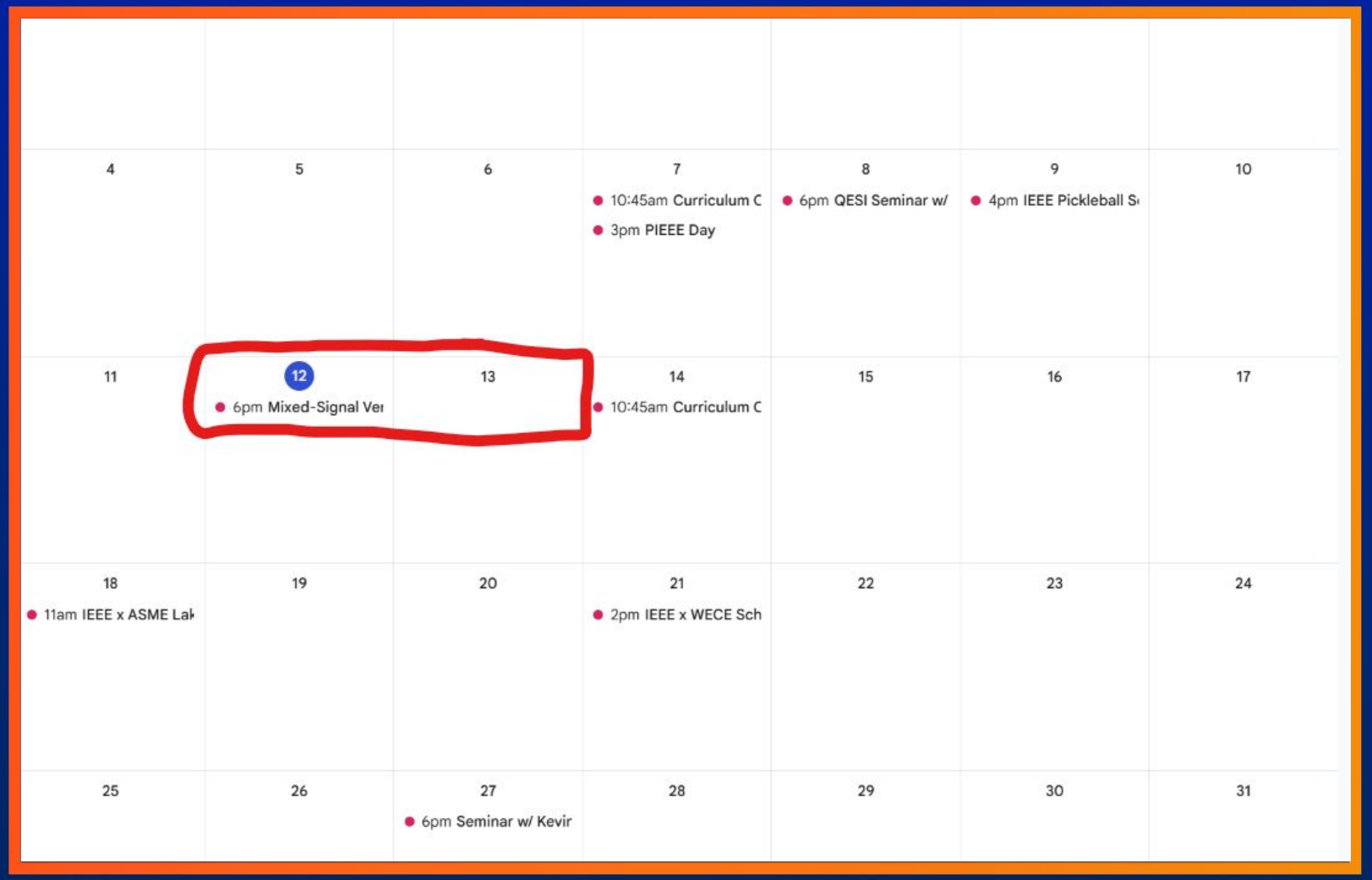


- Why the resistor? It limits current to protect the LED. No resistor and the circuit becomes a short!
- Polarity matters: current must enter the LED anode (+) and leave the cathode (-)



Project 2: LED Circuit

In this project, we will be soldering our own custom PCB made for an awesome LED Keychain!



Latest with IEEE!

Check the current IEEE calendar for workshops, GBMs, volunteering, and project deadlines!

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